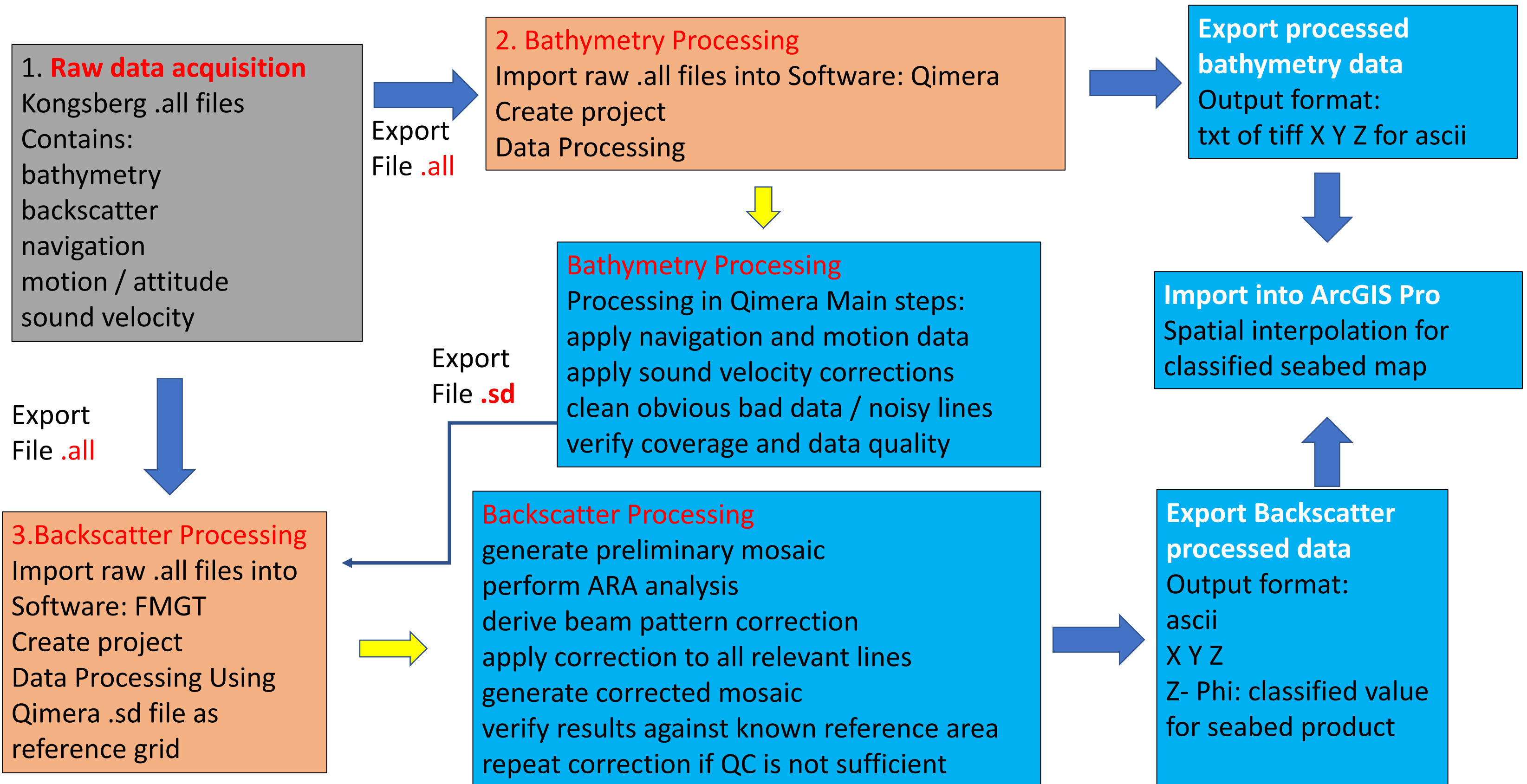


# Simple workflow scheme

Kongsberg **.all** Qimera **.sd** → FMGT → ArcGIS Pro

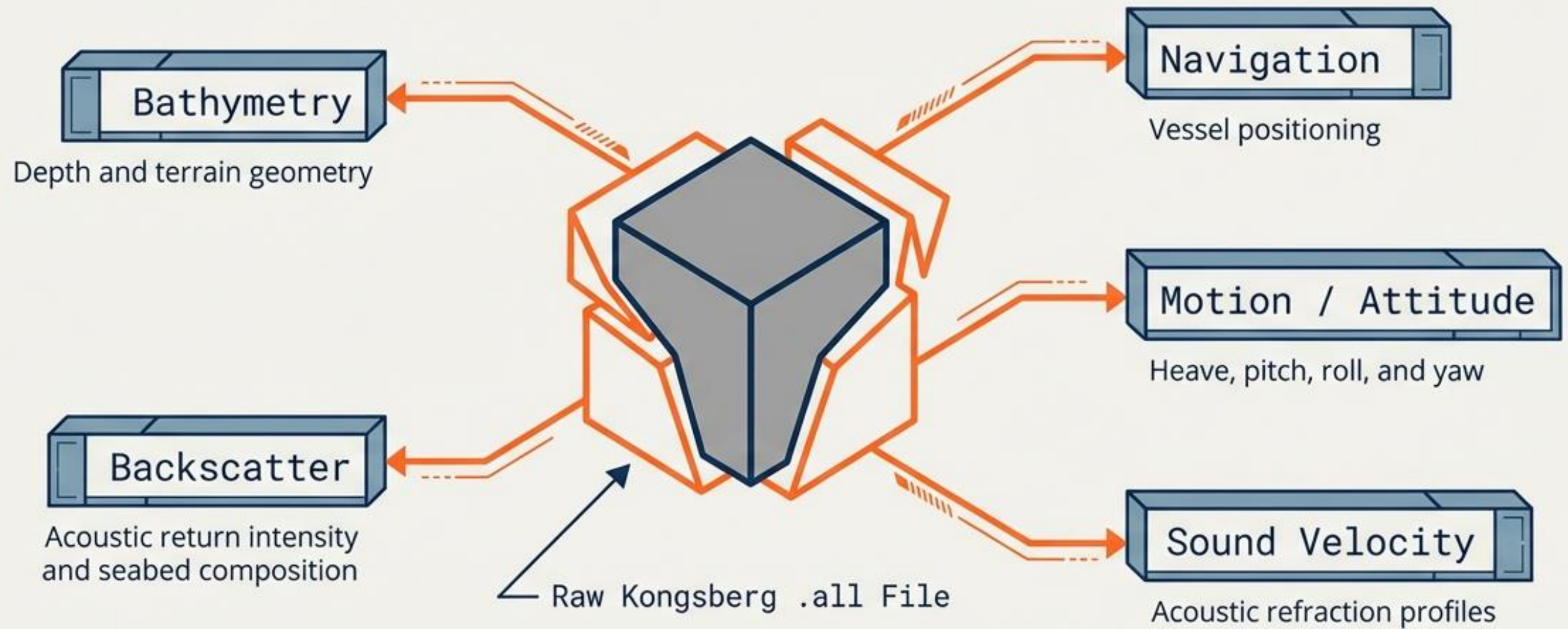


# Acoustic Data Processing Pipeline

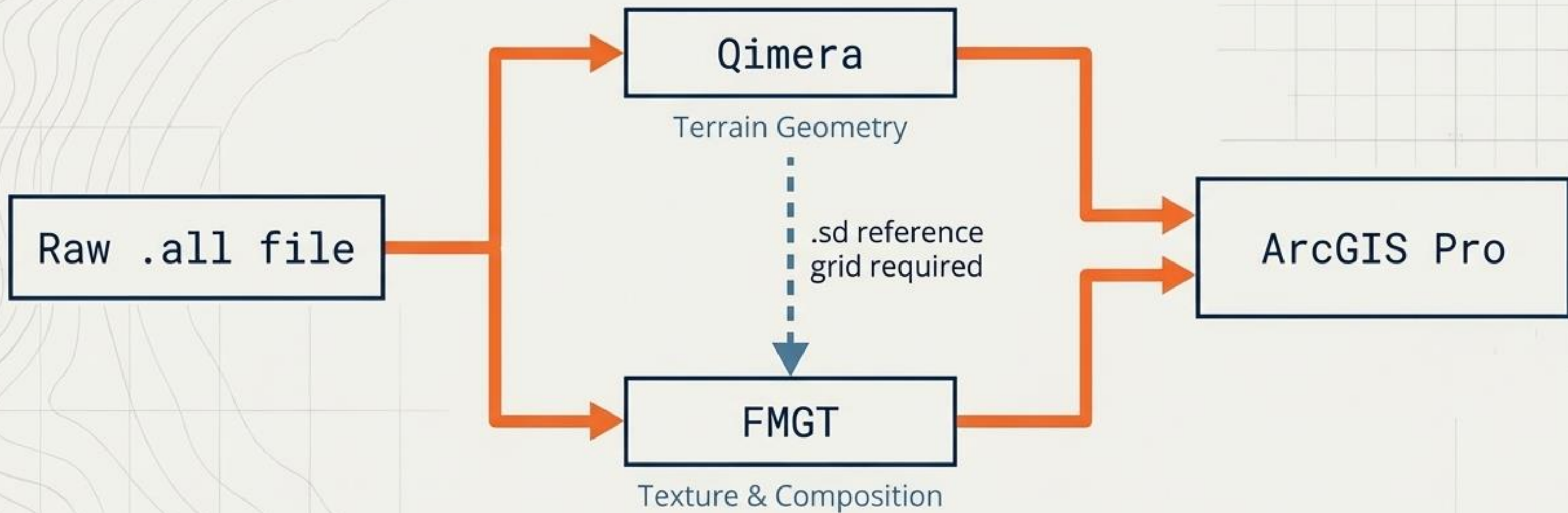
Standardising the journey from raw Kongsberg .all files to classified spatial seabed models.



# Deconstructing the Kongsberg .all File

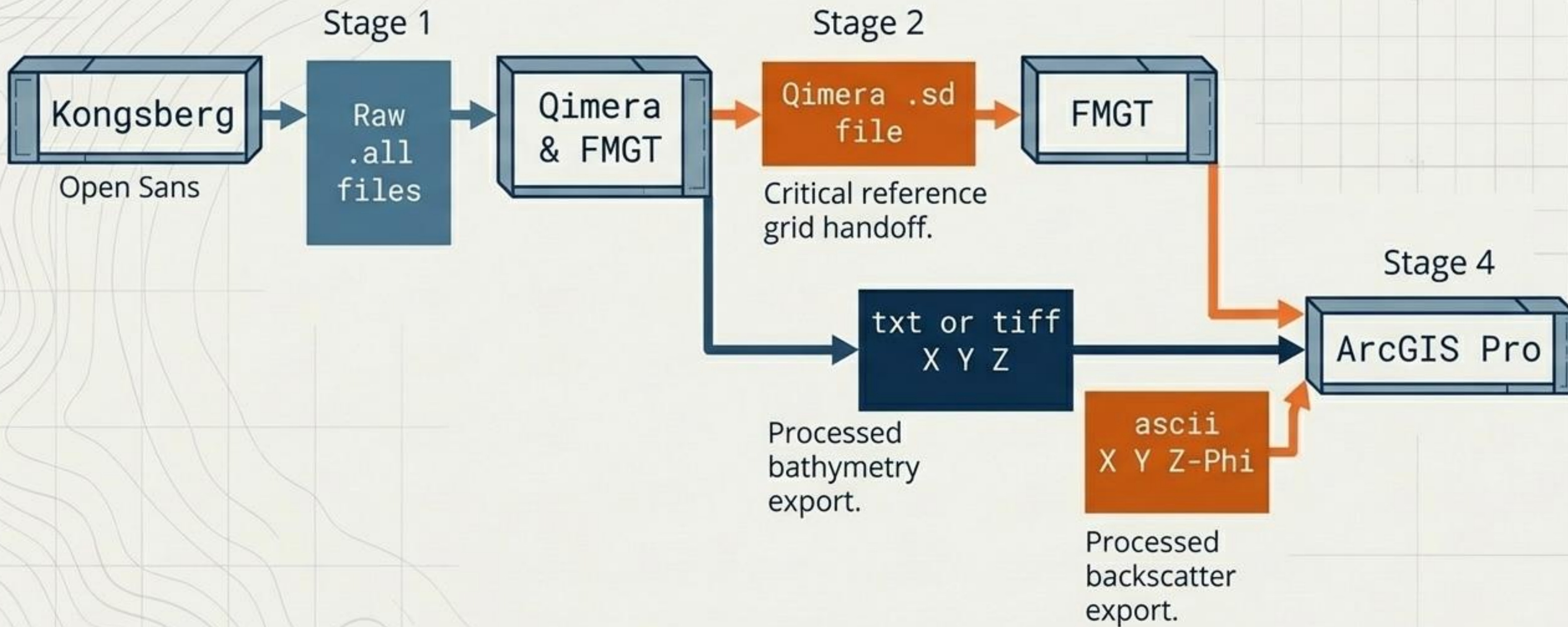


# The Diverge-and-Converge Processing Arc





# Software Handoff Mapping



# The Processing Bifurcation Matrix

|                                | Bathymetry (Qimera)                         | Backscatter (FMGT)                            |
|--------------------------------|---------------------------------------------|-----------------------------------------------|
| <b>Primary Objective</b>       | Terrain and Depth Geometry                  | Seabed Composition and Texture                |
| <b>Key Corrections Applied</b> | Navigation, Motion/Attitude, Sound Velocity | Angular Response Analysis (ARA), Beam Pattern |
| <b>Reference Dependency</b>    | Independent processing                      | Requires external Qimera .sd grid             |
| <b>Final Output Format</b>     | txt / tiff X Y Z or ascii                   | ascii X Y Z-Phi                               |



# Qimera: Establishing Terrain Geometry

**Import raw .all files  
& Create Project**



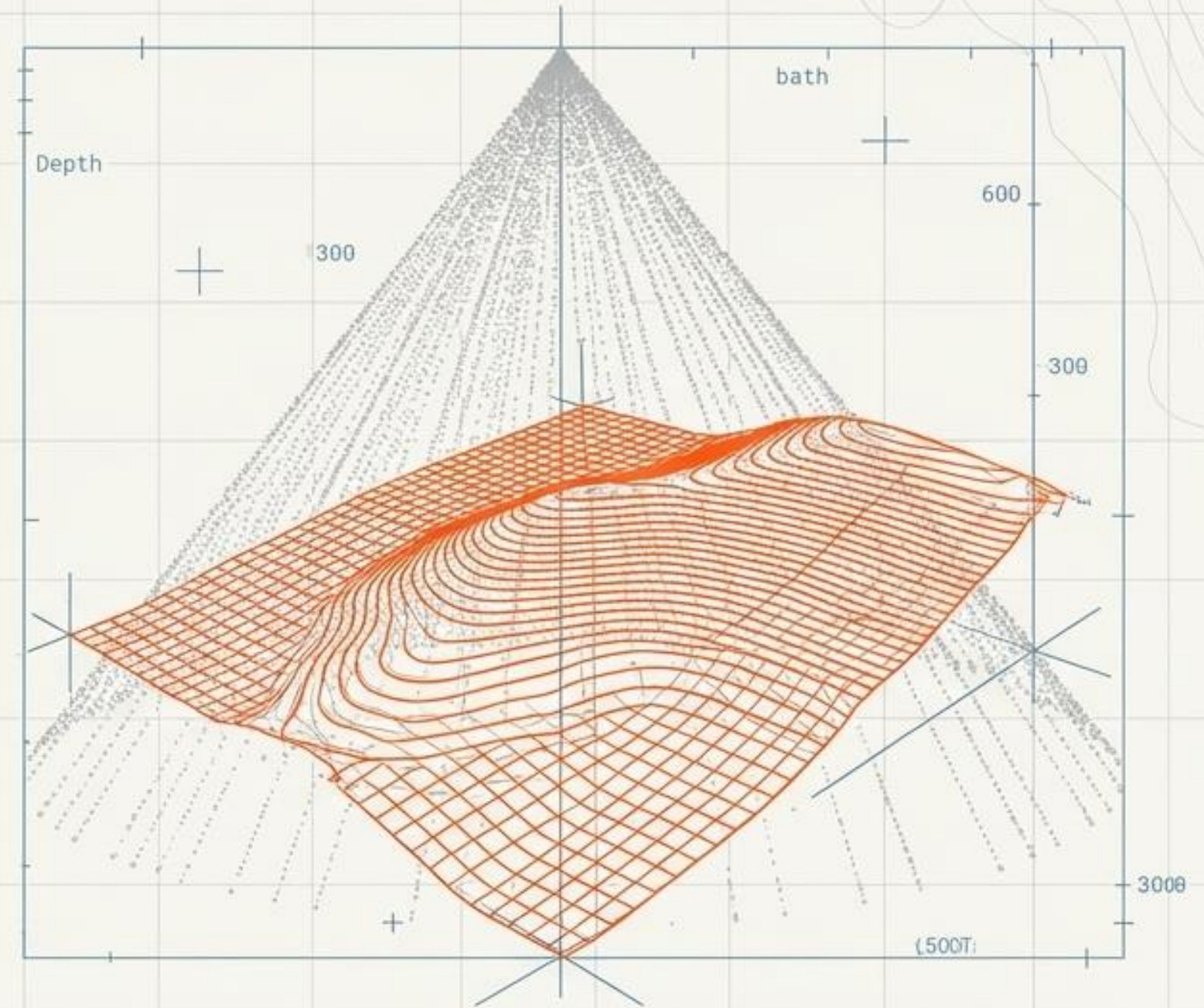
**Apply Navigation & Motion**

Correcting for vessel heave,  
pitch, roll, and positioning.



**Apply Sound Velocity**

Adjusting for acoustic refraction  
through the water column.



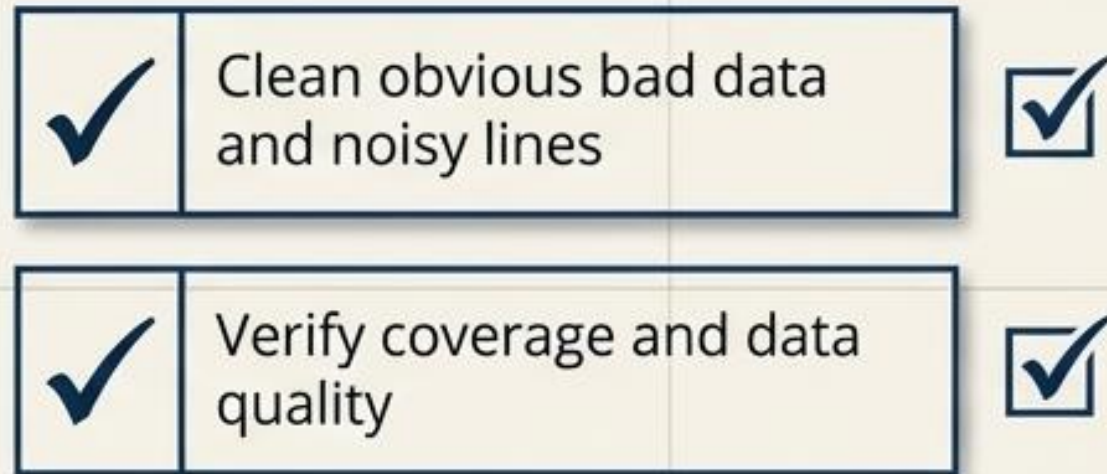


# Qimera: Quality Control and Dual-Export

Univers LT Std

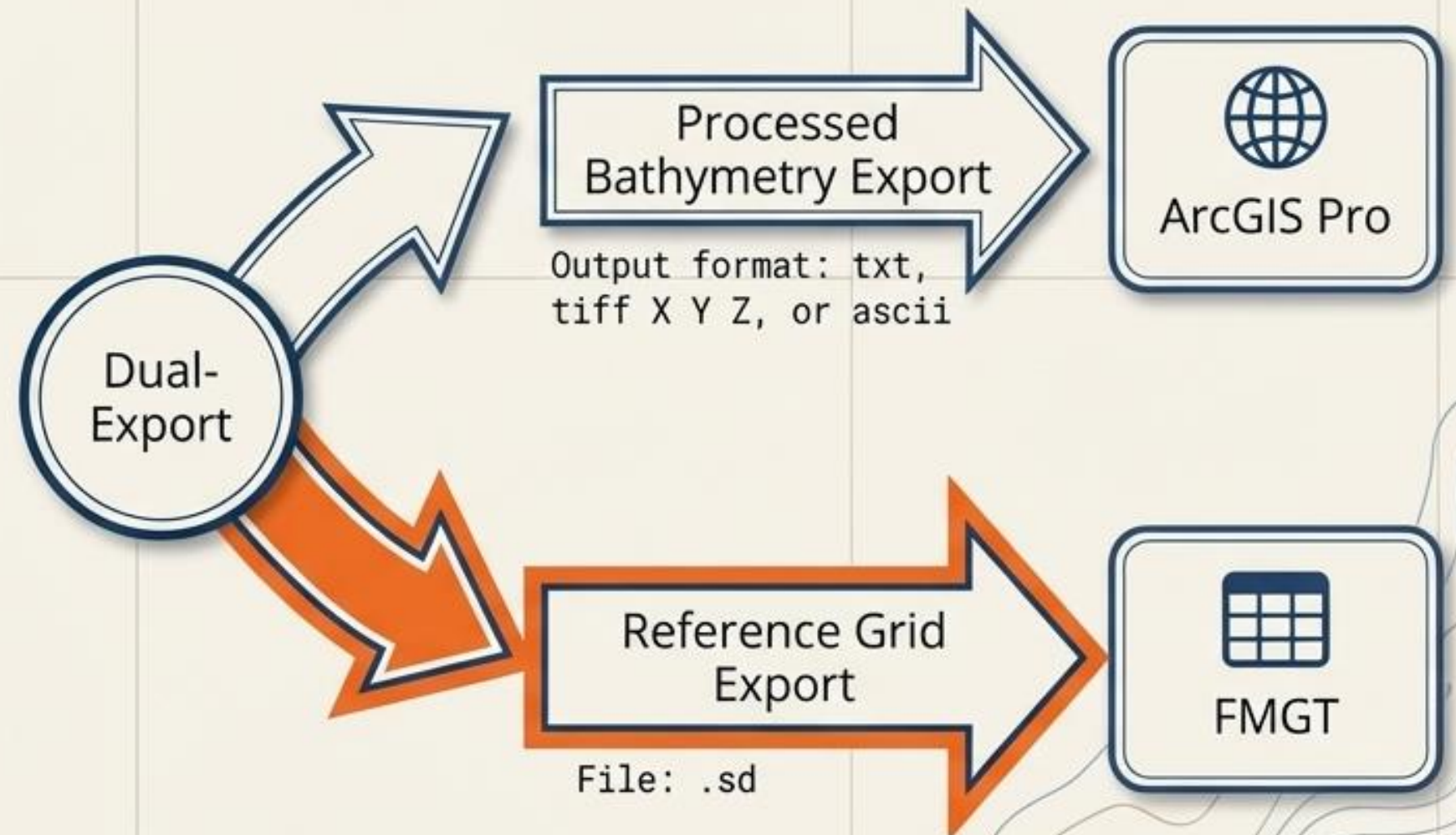
## Data Cleaning & Verification

Univers LT Std



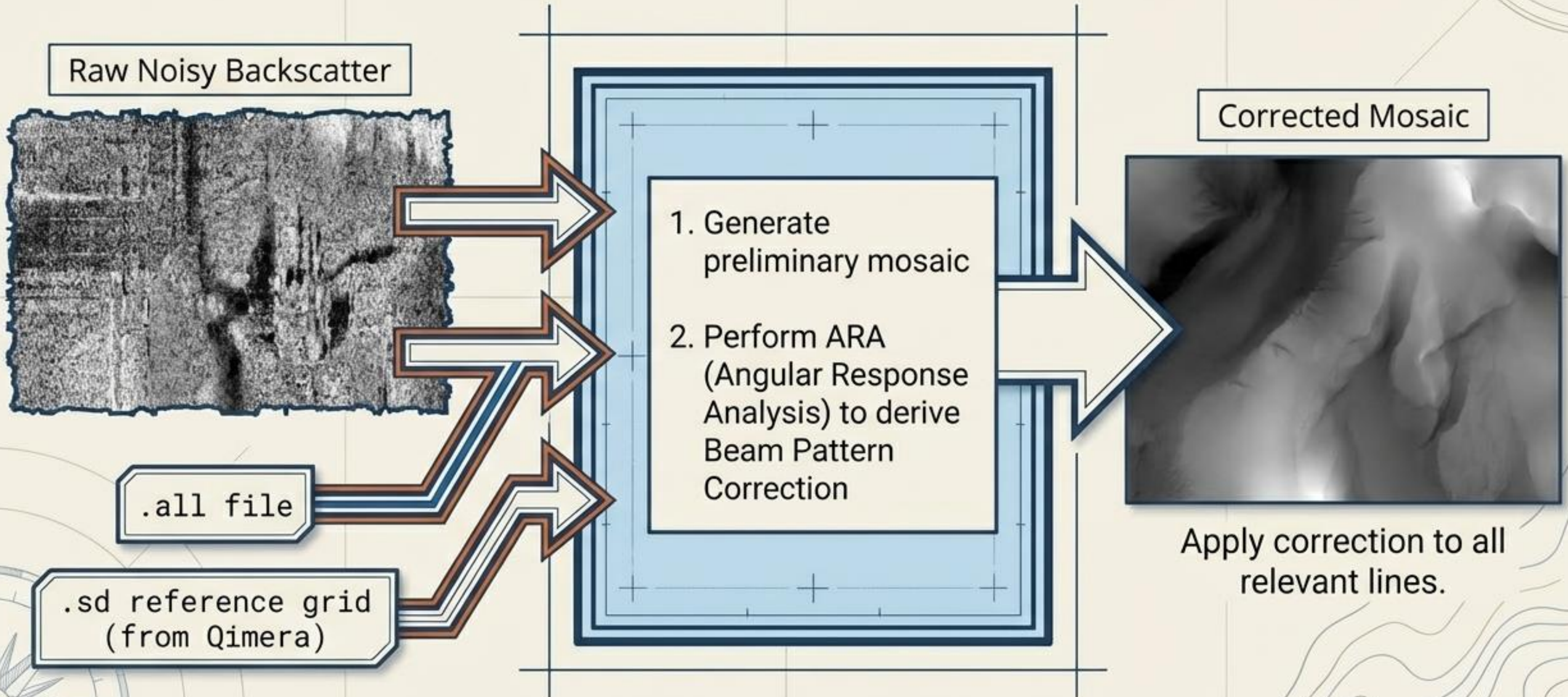
## Export Routing

Univers LT Std



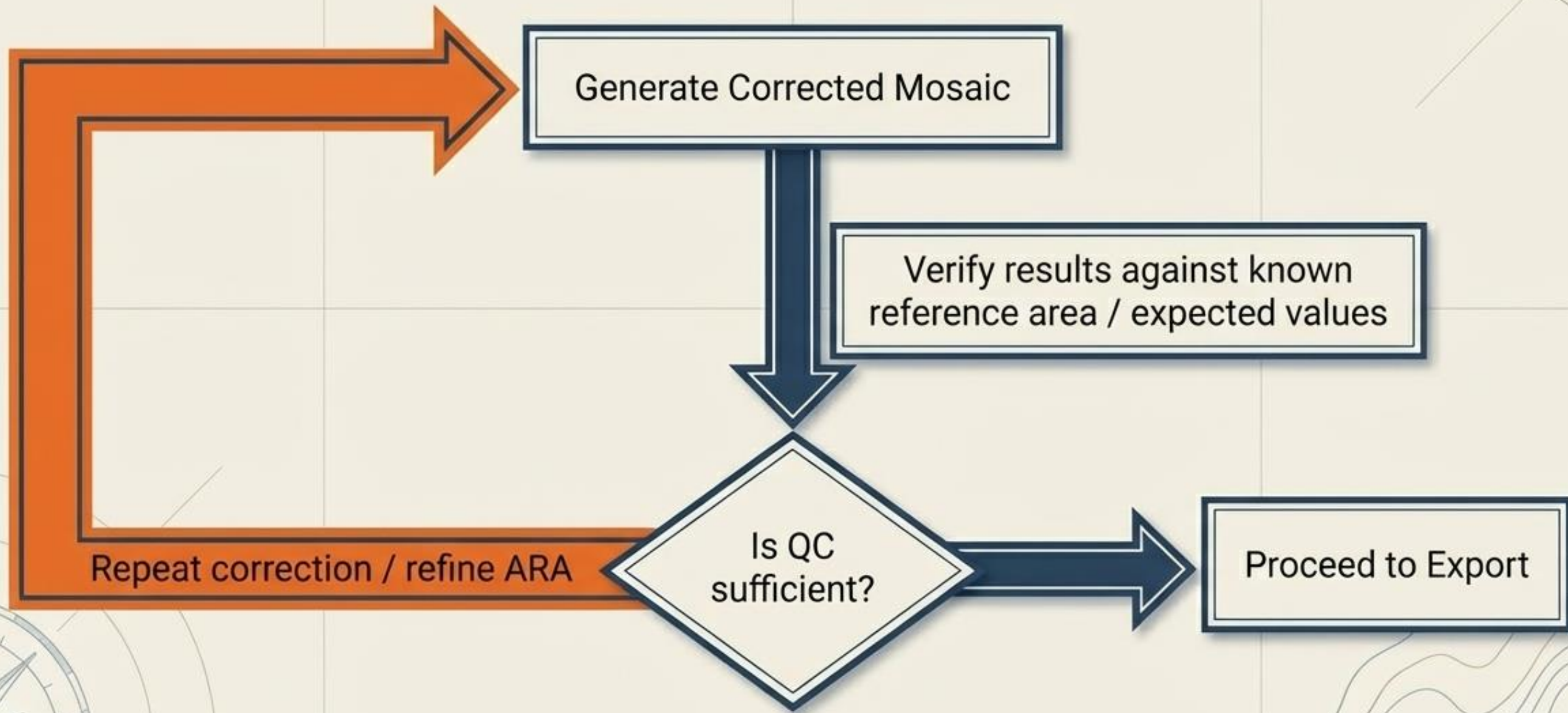


# FMGT: The Backscatter Engine





# The Backscatter QC Loop





# FMGT Export: Defining the Z-Phi Variable

**ascii [X][Y][Z][Phi]**

**X & Y**

Spatial Coordinates

**Z**

Depth Geometry

**Phi**

The classified value for the seabed product. Merges spatial location with derived acoustic classification.



# Synthesis: Recombining Terrain and Texture

Bathymetry Geometry  
(txt/tiff X Y Z from Qimera)

Backscatter Classification  
(ascii X Y Z-Phi from FMGT)



Import into ArcGIS Pro



# Finalizing the Spatial Seabed Map

## Process

Spatial interpolation

## Action

Translating discrete point cloud and text grid data (X Y Z-Phi) into continuous raster surfaces.

## Output

The definitive Classified Seabed Map.

